THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

May 23, 2025 (Friday) 9:30 AM – 12:30 PM (3 hours)

Answer ALL **5 (FIVE)** questions from Section A and B.
All questions carry equal marks.
Marks for sub-questions (if any) are given.

Use a separate answer book for each section.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

Q1 (a) A hollow circular tube having an inside diameter of 60 mm and a wall thickness of 10 mm is subjected to a torque T=136kN×m. The elastic modulus is 200GPa and Poisson ratio is 0.3. One end is fixed, and another end is free. The total length is 1 m. Note the thickness is not very small compared to the diameter and shear stress is not uniform.

(i) Determine the maximum shear stress. **(5 marks)**

(ii) Determine the angle of twist of the tube at the free end. **(5 marks)**

(b) A frame has hinge connection at C and two hinge supports at A and E. Height of column AB is 5m and length of beam BD is 10m. The load is applied at CD part and AB part as shown in Figure 1(b).

(i) Draw the shear force diagram (SFD) of the frame with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. Note that SFD of all components need to be plotted. **(5 marks)**

(ii) Draw the bending moment diagram (BDM) of the frame with pertinent values. Noe that bending moment should be drawn on the **tensile** side. Note that BMD of all components need to be plotted. **(5 marks)**

Q2 (a) A cantilever metal beam in Figure 2(a) has the section as shown in Figure 2(b). The total height is 500 mm and total width is 200 mm. The web thickness is 1 mm. The top and bottom flange plates have 200 mm width and 10 mm thickness. The density of the steel material is 7850kg/m^3. An additional axial tension force of 30kN is applied at free end. Gravity acceleration is 9.8m/s^2. The total span length is 30 m. Consider self-weight of the steel beam and axial tension force and neglect buckling for this beam. The yield strength is obtained from the tensile test as 165 MPa. Safety factor is 1.0.

(i) Check the maximum normal stress and maximum shear stress for the beam. **(5 marks)**

(ii) Check the safety of the beam using Tresca criterion. **(5 marks)**

(b) A point is subject to plane stress state as shown in Figure 2(c). x axis is horizontal and pointing to the right. y axis is vertical. The von Mises criterion is used for the material. The yield strength is obtained from the tensile test as 140 MPa. Safety factor is 1.0.

(i) Determine the stress acting on an element oriented at an angle θ=30 degrees from the x axis. Rotating counterclockwise from x axis is positive. **(5 marks)**

(ii) Check the safety of the point using von Mises criterion. **(5 marks)**

Q3 A steel tube AB of hollow circular cross section has a sliding support at the base and is pinned at the top to a horizontal beam supporting a load Q. The horizontal beam is rigid. The outside diameter is 200 mm and thickness is 10 mm. Safety factor of 1 for this problem. Elastic modulus E=200GPa. Check safety of the tube.

SECTION B

FIGURES

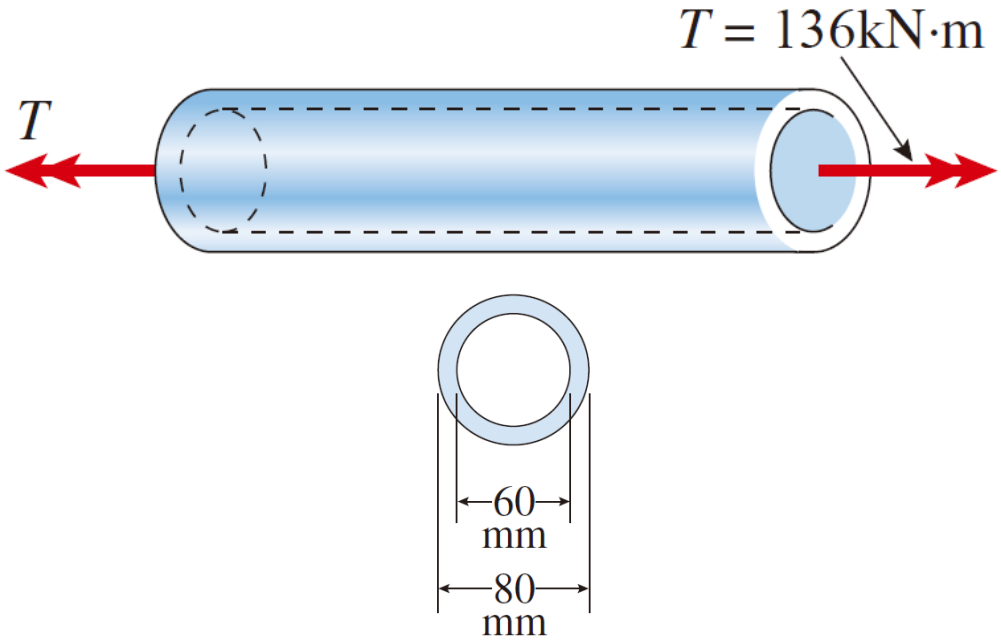


Figure 1(a). Beam

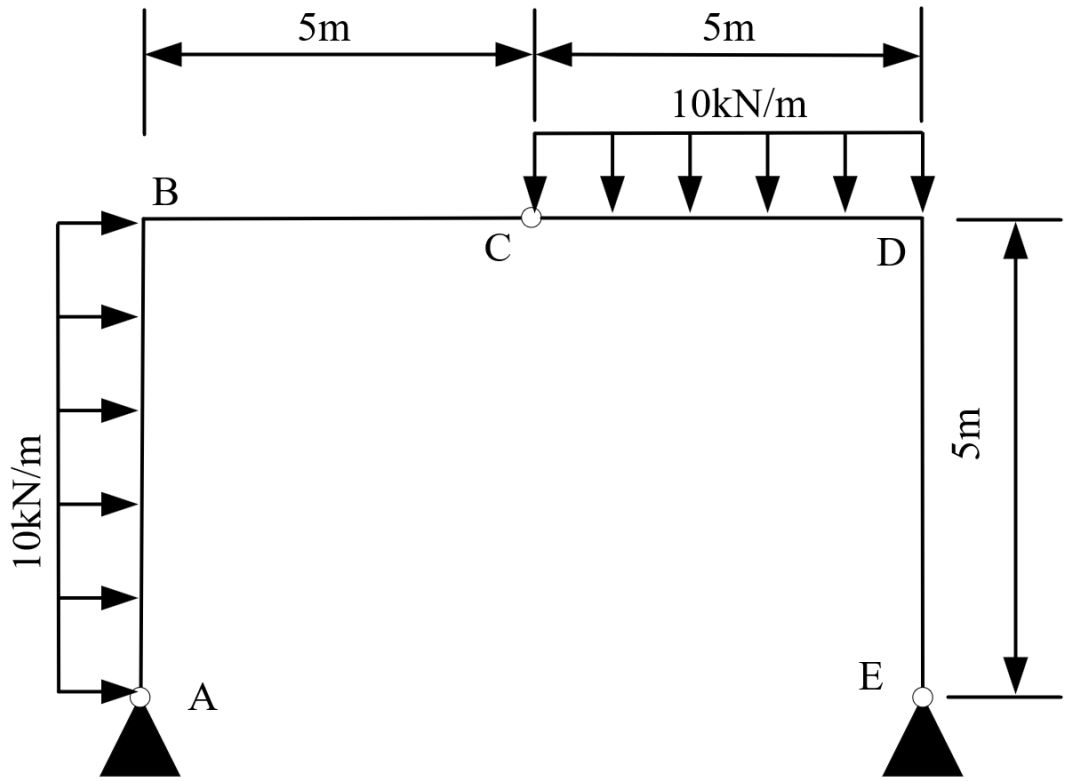


Figure 1(b). Plane frame

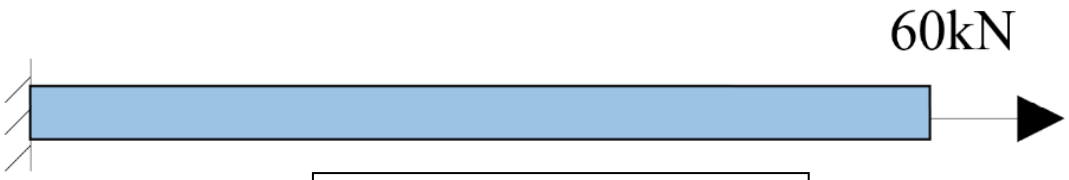


Figure 2(a). Cantilever beam

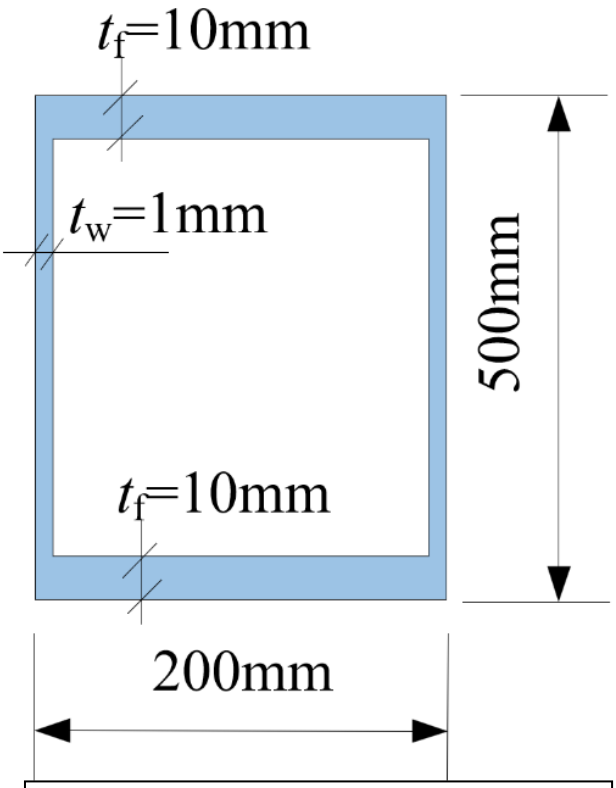


Figure 2(b). Simple beam section

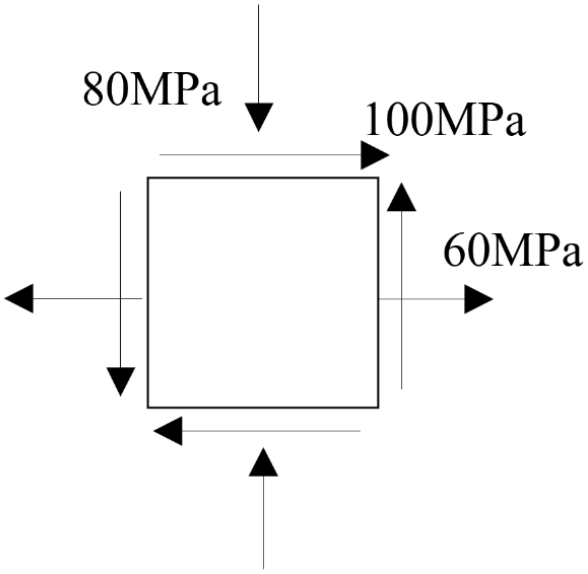


Figure 2(c). Plane stress at a point

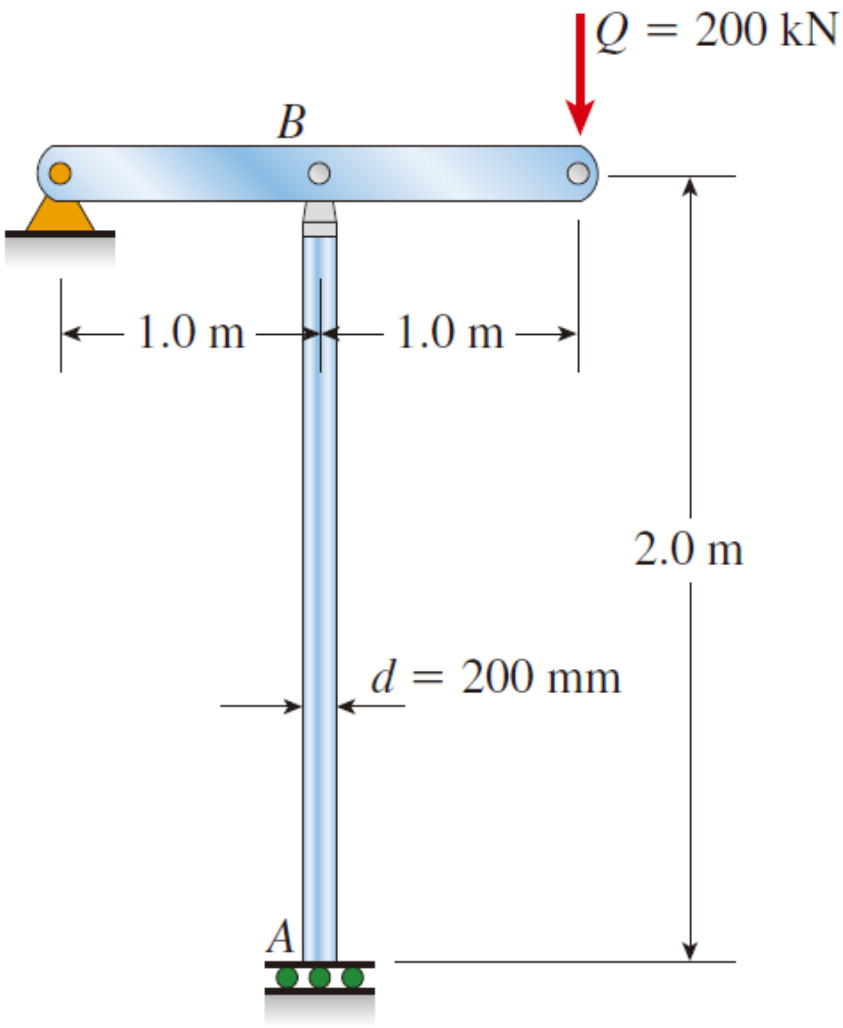


Figure 3. Column

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